

Department of Mechanical Engineering
University of South Carolina
Columbia, SC 29208
January 30, 2026

Dear Members of the Evaluation Committee:

I recommend Mateo Altomare for the SC Space Grant Undergraduate Research Award. As his faculty advisor in the Integrated Multiphysics and Systems Engineering Laboratory (iMSEL) since June 2025, I have directly supervised his research on multi-spectral sensor fusion for autonomous maritime perception and can speak specifically to his capabilities and readiness for the proposed project.

Mateo joined iMSEL as a rising sophomore with no prior experience in machine learning or computer vision. He progressed rapidly from introductory Python to implementing camera calibration routines for our thermal-visible sensor rig, and was independently writing PyTorch training scripts for our Detection Transformer (DETR) model on maritime imagery within his first semester. He mastered transfer learning through independent study, applying pretrained networks to thermal imagery—a task that required him to bridge two distinct imaging domains. He is now co-first-author on a paper submitted to ASME IDETC 2026 presenting a method for pixel-level infrared-to-visible image registration—a contribution that demanded graduate-level understanding of geometric transformations and multi-modal coordinate alignment, which he demonstrated by independently deriving and implementing the registration pipeline end-to-end.

Three accomplishments illustrate his technical range. First, he deployed real-time object detection on a Raspberry Pi for automated target identification in marine radar displays, optimizing a deep learning model to run on an edge device with no GPU. Second, he built our multi-spectral detection pipeline that fuses thermal and visible imagery for cross-modal calibration—this pipeline now generates all training data for his proposed framework. Third, he curated and annotated approximately 750 thermal-visible frame pairs spanning 1–75 m depth range, designing a structured labeling protocol that tracks scene conditions, object depth, and registration quality for each pair.

His proposed research—Learned Depth-Adaptive Registration (LDAR)—addresses a real limitation in our current workflow: our existing registration method requires knowing object depth in advance. Mateo’s proposal would automate depth estimation and eliminate the need for manual specification and additional hardware. I have reviewed his proposed architecture and training plan in detail; the approach is technically sound, the dataset is sufficient for the method, and the accuracy target is ambitious but achievable given our preliminary results.

This work has direct relevance to NASA’s Artemis program. Lunar South Pole rovers will navigate permanently shadowed regions where visible cameras are ineffective and GPS does not exist. Thermal-visible sensor fusion is essential for obstacle detection in these environments, and the depth-adaptive registration techniques Mateo is developing address exactly this alignment problem for mass-constrained platforms.

Beyond research, Mateo co-leads the Gamecock Motorsports Formula SAE chassis team and maintains a 3.91 GPA. When early thermal calibration tests in August 2025 produced registration errors exceeding 10 pixels—far above our target—he traced the problem to a

temperature-dependent lens distortion effect, redesigned the calibration sequence to account for thermal drift, and reduced error below 3 pixels within two weeks. Based on his completion of all Fall 2025 milestones ahead of schedule, I expect him to execute the proposed research plan fully and deliver the stated outcomes.

I give Mateo Altomare my highest recommendation for this award.

Sincerely,

Dr. Yi Wang
Associate Professor
Department of Mechanical Engineering
University of South Carolina
ywang@sc.edu